
The Power of Vaccination From Smallpox to COVID-19 in Record Time

A Data-Driven Overview of Public Health Transformation (1796–2020)

SOURCE

Our World in Data

AUDIENCE

Public Health Professionals

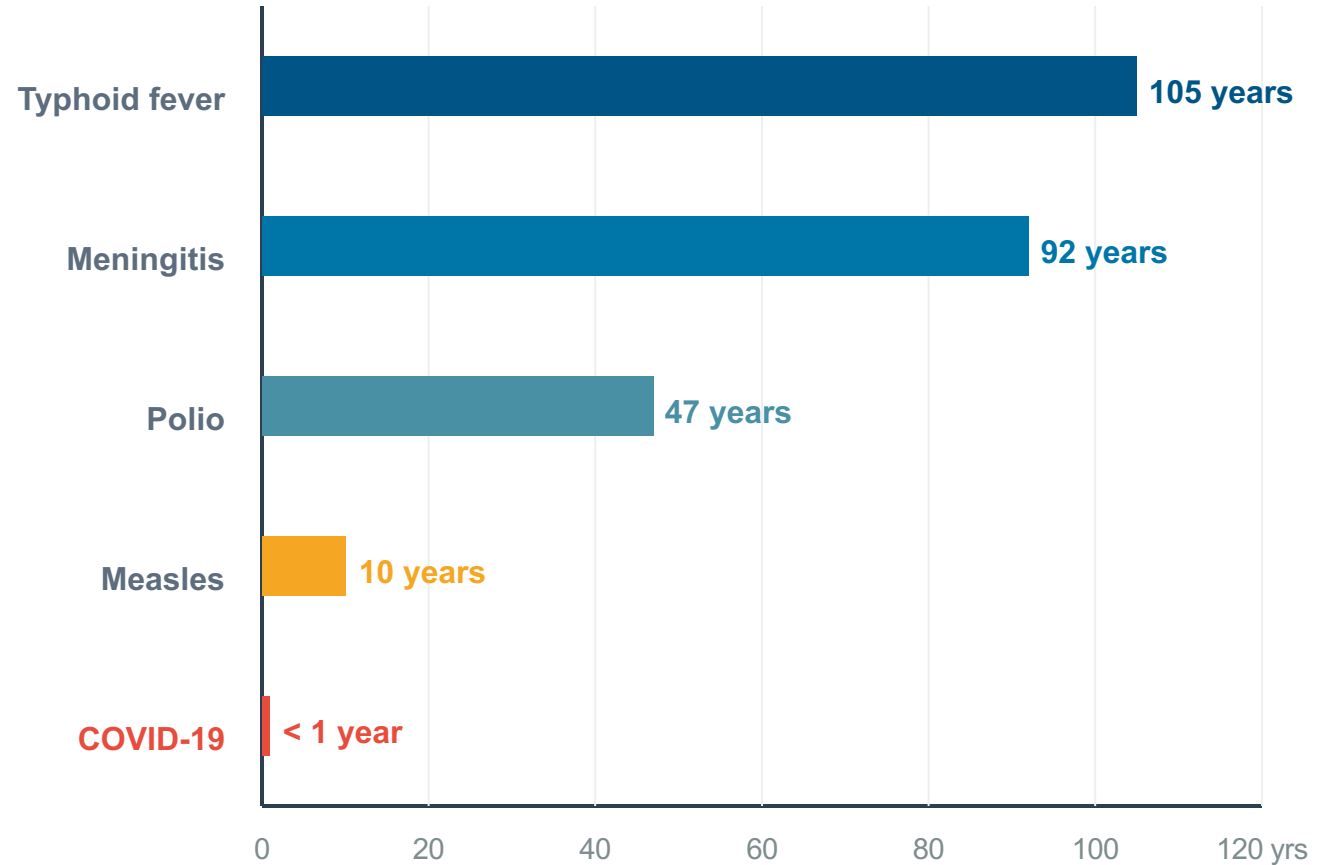
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From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Innovation in Development

- Early vaccines (e.g., Typhoid) took over a century from pathogen identification to licensure.
- Advancements in modern tools like advanced microscopy, genome sequencing, and mRNA technology enabled faster design.
- **COVID-19 set a record timeline: Under 1 year**

Years from Pathogen Identification to Vaccine Licensure



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the

Disease	Pre-Vaccine Cases (per million/yr)	Post-Vaccine Cases (per million/yr)	Case Reduction	Pre-Vaccine Deaths (per million/yr)	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

● 100% or near 100% Elimination ● Remaining Burden (Incomplete Impact)

Measles Cases Collapsed Across All 50 US States After Vaccine Mandat

Policy Milestones

Before 1963: High Incidence

High rates (100–1,000+ cases per 100,000) across nearly all states.

1963 & 1971: First Vaccines

First measles vaccine (1963) and MMR vaccine (1971) introduced, leading to initial steep declines.

1980: Kindergarten Mandate

Cases dropped to near zero across all states after school entry vaccination became mandatory.

Vaccines reduced measles cases across US states



Data source: Project Tycho (2018); Centers for Disease Control and Prevention (1959–2022)
OurWorldinData.org — Research and data to make progress against the world's largest problems.

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Smallpox: The Only Human Disease Eradicated Through Vaccination

Global Eradication Campaign Timeline



Why Was Smallpox Uniquely Suited for Eradication?

- ✓ **Human-Only Host:** The virus infected only humans, meaning there were no animal reservoirs to sustain it.
- ✓ **Distinguishable Symptoms:** Infections were visually obvious, enabling easy identification and isolation.
- ✓ **Effective & Cheap Vaccine:** The vaccine provided strong, long-lasting immunity and was highly cost-effective.

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

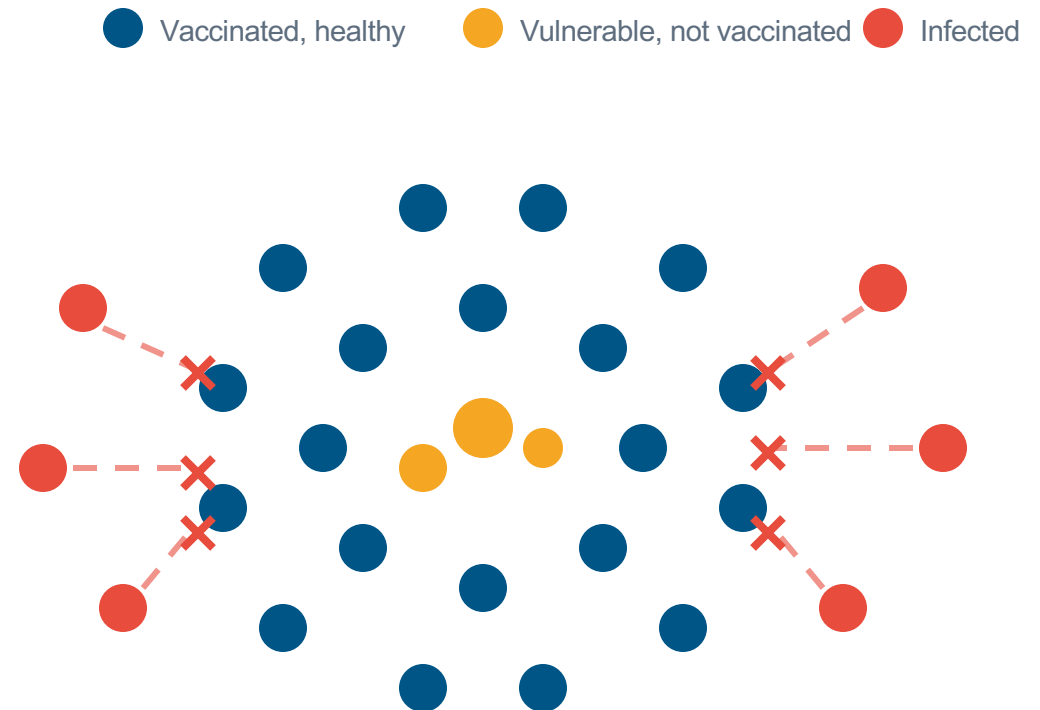
The Shield Effect

When a critical portion of a community is immunized, most members are protected against that disease because there is little opportunity for an outbreak.

Who Does It Protect?

- ✓ Newborns too young to be vaccinated
- ✓ Immunocompromised individuals
- ✓ Patients undergoing chemotherapy
- ✓ The elderly with weakened immune systems

Community Protection Model



Pathogens are unable to reach the vulnerable population

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Incomplete Disease Control

Pneumococcal Disease

Case Reduction:**40.5%** | Death Reduction:**31.3%**

Hepatitis A

Case Reduction:**89.0%** | Death Reduction:**88.7%**

Acute Hepatitis B

Case Reduction:**83.9%** | Death Reduction:**83.6%**

Barriers to Full Eradication



Access Gaps

Globally, not everyone has access to life-saving vaccines due to infrastructure and economic limits.



Supply Shortages

Intermittent supply shortages put children at risk in vulnerable regions worldwide.



Trust & Education

Poor scientific understanding and misinformation can significantly reduce vaccine uptake and trust.

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the J

Summary of Findings

- ✓ **100% Elimination Achieved:**
eliminated in the US, demonstrating the profound power of vaccination.
- ✓ **Accelerated Development:**
(typhoid) to under 1 year (COVID-19).
- ✓ **Herd Immunity Protection:**
the most vulnerable members of society.
- ✓ **Persisting Coverage Gaps:**
hepatitis due to barriers in access, supply, and public trust.

Call to Action

Global health systems must prioritize improving equitable access, ensuring robust supply chains, and fostering public trust to close remaining coverage gaps and conquer remaining diseases.